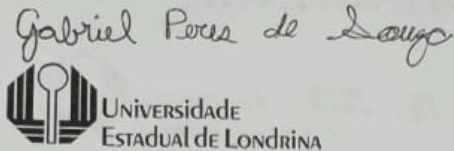


Exercícios Práticos Compiladores - Entrega 1

Gabriel Peres de Souza



1COP020/2COP048 - Lista de Exercícios 01

1. **Exercício Teórico:** Determine os seguintes fechos de Kleene:

- (a) $\{ab, c\}^* = \{\lambda, ab, c, abc, cab, abab, cc, ababab, ccc, \dots\}$
 (b) $\{a, b, c\}^+ = \{a, b, c, aa, bb, cc, ab, ac, ba, bc, ca, cb, aaa, bbb, \dots\}$
 (c) $\{abc\}^* = \{\lambda, abc, abcabc, abcabcabc, abcabcabcabc, \dots\}$

2. **Exercício Teórico:** Determine o resultado das seguintes operações nas linguagens indicadas:

- (a) $\{a, b\} \cup \{c, d\} = \{a, b, c, d\}$
 (b) $\{100, 010, 110\} \cup \{00, 01, 11\} = \{100, 010, 110, 00, 01, 11\}$
 (c) $\{101, 110\} \cdot \{00, 11\} = \{10100, 10111, 11000, 11011\}$
 (d) $\{Anab, Emanu, Gabri, Isab, Rafa\} \cdot \{ela\} = \{Anabela, Emanuela, Gabriela, Isabela, Rafaela\}$

3. **Exercício Teórico:** Considere o seguinte alfabeto:

$$\Sigma = \{j, l, n\}$$

determine:

- (a) $\Sigma^2 = \{jj, jl, jn, ll, ln, nn\}$
 (b) $\Sigma^3 = \{jjj, jjl, jln, jll, jln, jnn, llj, lln, lln, lln, lnn, nll, nln, nnn\}$
 (c) $\Sigma^2 \cup \Sigma^2 = \Sigma^2$
 (d) $\Sigma^2 \cdot \Sigma^2 = \Sigma^4$

4. **Exercício Teórico:** Considere os seguintes alfabetos:

$$\Sigma_1 = \{H_2, Fe\}$$

$$\Sigma_2 = \{SO_4, O_2, Cl_2\}$$

determine:

- (a) $\Sigma_1^2 = \{H_2H_2, H_2Fe, FeH_2, FeFe\}$
 (b) $\Sigma_1^2 \cup \Sigma_2^2 = \Sigma_1^2 \cup \Sigma_2^2$
 (c) $\Sigma_1^1 \cdot \Sigma_2^1 = \{H_2SO_4, H_2O_2, H_2Cl_2, FeSO_4, FeO_2, FeCl_2\}$

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Lista 2

LISTA 2 - EXERCÍCIO 1 - LETRA A

$$c^*ab^+(a|b|c)^*$$

LISTA 2 - EXERCÍCIO 1 - LETRA B

$$(aa)^*1b|c$$

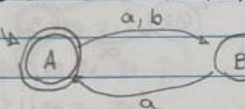
LISTA 2 - EXERCÍCIO 2 - LETRA A

O autômato reconhece todos números binários de 4 dígitos, de 0000 a 1111.

LISTA 2 - EXERCÍCIO 2 - LETRA B

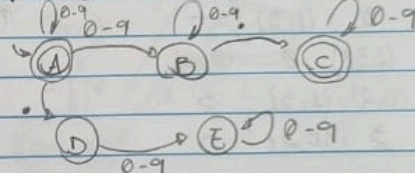
O autômato reconhece as palavras formadas por "a" repetidas $5x+1$ vezes, sendo x um valor natural qualquer.

LISTA 2 - EXERCÍCIO 3 - LETRA A

$$(a|b)a^* =$$


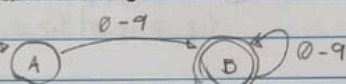
A finite automaton with two states: A (start) and B (final). Transitions: A to B on 'a, b'; B to A on 'a'.

LISTA 2 - EXERCÍCIO 3 - LETRA D

$$([0-9]^+ \cdot [0-9]^+)^* | ([0-9]^+ \cdot [0-9]^+)^*$$


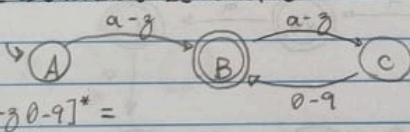
A finite automaton with five states: A (start), B, C, D, E. Transitions: A to B on '0-9'; B to C on '0-9'; C to A on '0-9'; A to D on '0-9'; D to E on '0-9'; E to A on '0-9'.

LISTA 2 - EXERCÍCIO 3 - LETRA B

$$[0-9]^+ =$$


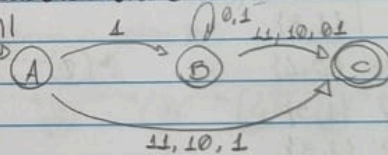
A finite automaton with two states: A (start) and B (final). Transition: A to B on '0-9'; B to B on '0-9'.

LISTA 2 - EXERCÍCIO 3 - LETRA C

$$[a-z][a-z0-9]^* =$$


A finite automaton with three states: A (start), B (final), C. Transitions: A to B on 'a-z'; B to C on 'a-z'; C to B on '0-9'.

LISTA 2 - EXERCÍCIO 3 - LETRA E

$$(1|0|1)^+ (1|1|0|0|1)^+ | (1|1|1|0|0|1)^+$$


A finite automaton with three states: A (start), B, C (final). Transitions: A to B on '1'; B to C on '0, 1'; C to A on '1, 0, 1'; A to C on '1, 1, 0, 1'.

LISTA 2 - EXERCÍCIO 4 - LETRA A

$$(1.f)$$

LISTA 2 - EXERCÍCIO 4 - LETRA B

$$(a-z)((a-z)|(0-9))^*$$

LISTA 2 - EXERCÍCIO 4 - LETRA D

$$((0-9)^+ \cdot (0-9)^*) | ((0-9)^+)$$

LISTA 2 - EXERCÍCIO 4 - LETRA E

$$--(a-z)^* | (bank)^+$$

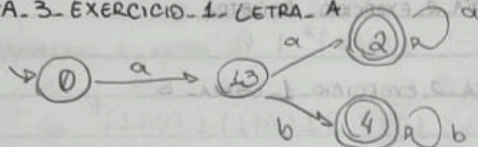
LISTA 2 - EXERCÍCIO 4 - LETRA C

$$(0-9)^+$$

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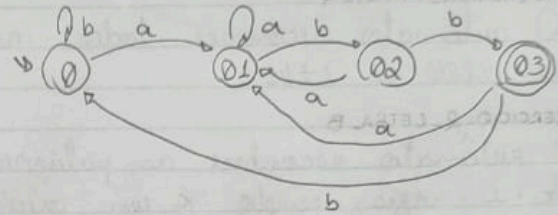
Lista 3

LISTA-3-EXERCICIO-1-LETRA-A



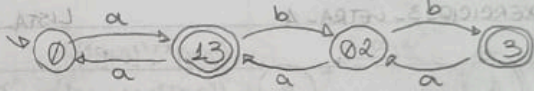
LISTA-3-EXERCICIO-1-LETRA-B

	a	b
0	{0,1}	0
{0,1}	{0,1}	{0,2}
{0,2}	{0,1}	{0,3}
{0,3}	{0,1}	0



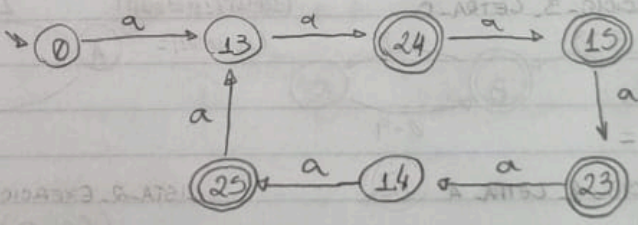
LISTA-3-EXERCICIO-1-LETRA-C

	a	b
0	{1,3}	-
{1,3}	0	{0,2}
{0,2}	{1,3}	3
3	{0,2}	-



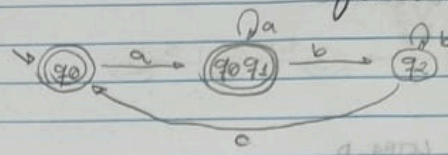
LISTA-3-EXERCICIO-1-LETRA-D

	a
0	{1,3}
{1,3}	{2,4}
{2,4}	{1,5}
{1,5}	{2,3}
{2,3}	{1,4}
{1,4}	{2,5}
{2,5}	{1,3}



LISTA 3 - EXERCICIO 1 - LETRA E

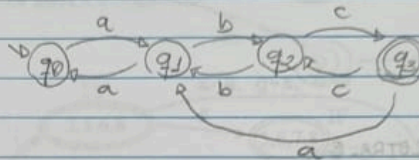
	a	b	c
q0	{q0, q1}	-	-
{q0, q1}	{q0, q1}	q2	-
q2	-	q2	q0



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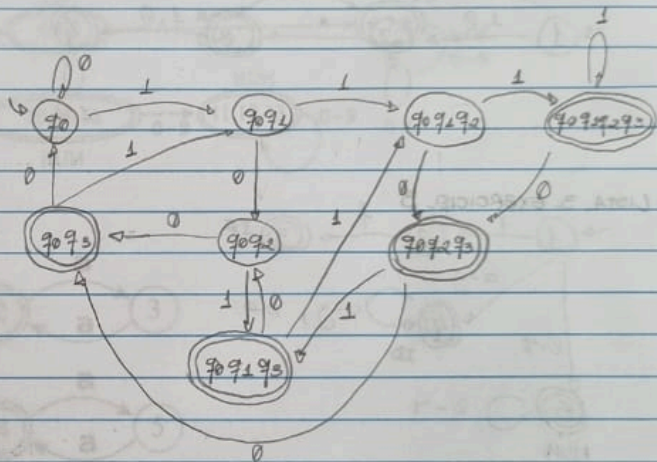
LISTA 3 - EXERCICIO 1 - LETRA F

	a	b	c
q0	q1	-	-
q1	q0	q2	-
q2	-	q1	q3
q3	q1	-	q2

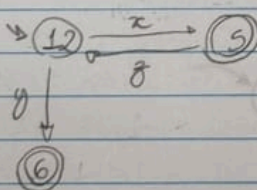


LISTA 3 - EXERCICIO 1 - LETRA G

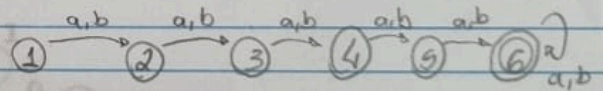
	0	1
q0	q0	{q0, q1}
{q0, q1}	{q0, q2}	{q0, q1, q2}
{q0, q1}	{q0, q2}	{q0, q1, q2}
{q0, q2}	q0	{q0, q1}
{q0, q1, q2}	{q0, q2, q3}	{q0, q1, q2, q3}
{q0, q1, q2}	{q0, q2}	{q0, q1, q2}
{q0, q2, q3}	{q0, q3}	{q0, q1, q3}
{q0, q1, q2, q3}	{q0, q2, q3}	{q0, q1, q2, q3}



LISTA 3 - EXERCICIO 2 - LETRA A



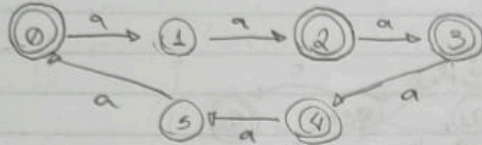
LISTA 3 - EXERCICIO 2 - LETRA B



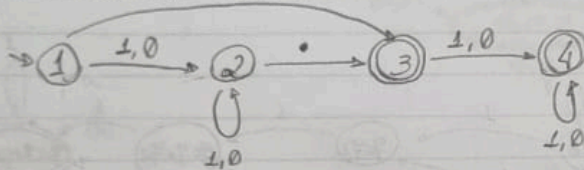
LISTA 3 - EXERCÍCIO 2 - LETRA C



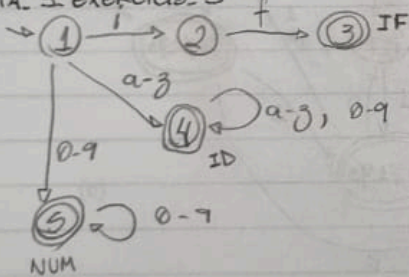
LISTA 3 - EXERCÍCIO 2 - LETRA D



LISTA 3 - EXERCÍCIO 2 - LETRA E



LISTA 3 - EXERCÍCIO 3



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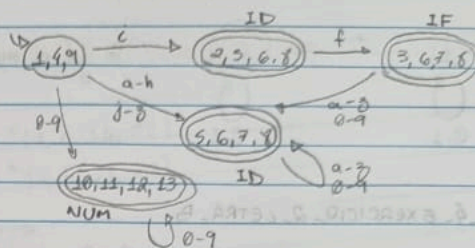
Lista 4

Gabriel Pires de Souza

LISTA-4-EXERCICIO-1-LETRA-A

* 2, 5, 6, 8	X						
* 2, 6, 8	X	X					
* 10, 11, 13	X	X	X				
* 3, 6, 7, 8	X	X	X		X		
* 6, 7, 8	X	X			X	X	
* 11, 12, 13	X	X	X			X	X
1, 4, 9	2, 5, 6, 8	5, 6, 8	10, 11, 13	3, 6, 7, 8	6, 7, 8		

Autômato minimizado:



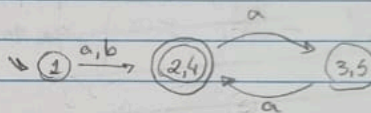
LISTA 4 - EXERCÍCIO 1 - LETRA B

*2	x			
3	x	x		
*4	x		x	
5	x	x		x
	1	*2	3	*4

a	b
---	---

autômato minimizado:

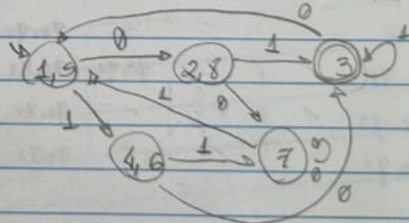
1	2	4
*2	3	-
3	2	-
*4	5	-
5	4	-



LISTA 4 - EXERCÍCIO 1 - LETRA C

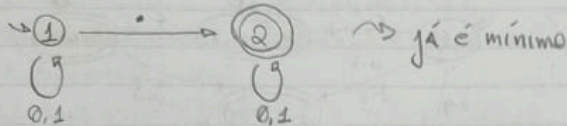
	0	1												
1	2	6	2	X										
2	7	3	*3	X	X									
*3	1	3	4	X	X	X								
4	3	7	(5)		X	X	X							
5	8	6	6	X	X	X		X						
6	3	7	7	X	X	X	X	X	X					
7	7	9	Y	X		X	X	X	X	X				
8	7	3		(1)	(2)	*3	(4)	5	6	7				

autômato minimizado:



LISTA 4 - EXERCICIO 2 LETRA A

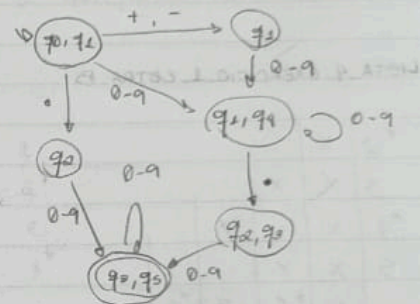
NFA-E convertido p DFA:



ER:
 $(011)^* \cdot (011)^*$

LISTA 4 - EXERCICIO 2 LETRA B

	+	-	0-9	.
q_0, q_1	q_1	q_1	q_1, q_4	q_2
q_1, q_4	-	-	q_1, q_4	q_2, q_3
q_2, q_3	-	-	q_3, q_5	-
q_3, q_5	-	-	q_3, q_5	-



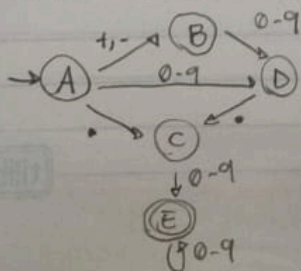
minimizando:

	+	-	.	0-9
q_0, q_1	q_1	q_1	q_2	q_1, q_4
q_1	-	-	-	q_1, q_4
q_2	-	-	-	q_3, q_5
q_1, q_4	-	-	-	q_3, q_5
q_2, q_3	-	-	-	q_3, q_5
q_3, q_5	-	-	-	q_3, q_5

	q_1	q_2	q_1, q_4	q_2, q_3	q_3, q_5
q_1	X	X	X	X	X
q_2	X	X	X	X	X
q_1, q_4	X	X	X	X	X
q_2, q_3	X	X	X	X	X
q_3, q_5	X	X	X	X	X

A = q_0, q_1
 B = q_1
 C = q_2, q_3, q_5
 D = q_1, q_4
 E = q_3, q_5

autômato minimizado:



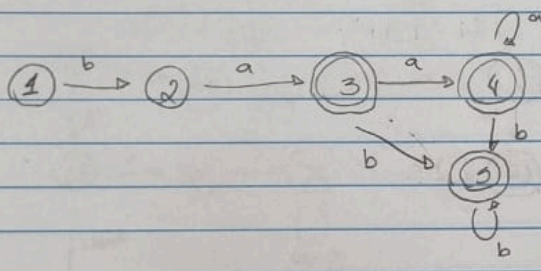
ER:
 $((+|-)0-9 | (0-9) | \lambda) \cdot (0-9)^+$

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LISTA 4 - EXERCICIO 3.

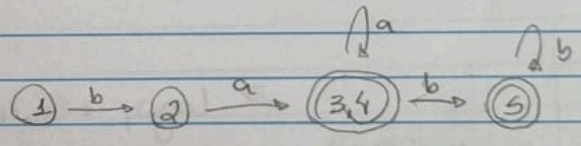
Autômato A:

	a	b
1 $\{16, 0, 17, 18\}$	-	$\{1, 2, 19, 20\}$
2 $\{1, 2, 19, 20\}$	* $\{0, 3, 17, 18, 21, 22, 14, 8, 15, 23\}$	-
3 $\{0, 3, 17, 18, 21, 22, 14, 8, 15\}$	$\{10, 4, 11, 12, 6, 13\}$	* $\{7, 6, 13, 23\}$
4 $\{10, 4, 11, 12, 6, 13\}$	* $\{7, 8, 15, 23, 5, 4, 11, 12, 6, 13\}$	* $\{7, 6, 13, 23\}$
5 $\{7, 8, 15, 23, 5, 4, 11, 12, 6, 13\}$	-	* $\{7, 6, 13, 23\}$



minimizandoo:

2	x			
* 3	x	x		
* 4	x	x	x	
* 5	x	x	x	x
	1	2	* 3	* 4

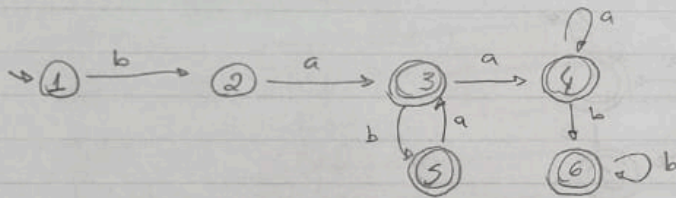


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LISTA 4 - EXERCÍCIO 3

Autômato B:

	a	b
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		
14		
15		
16		
17		



minimizando

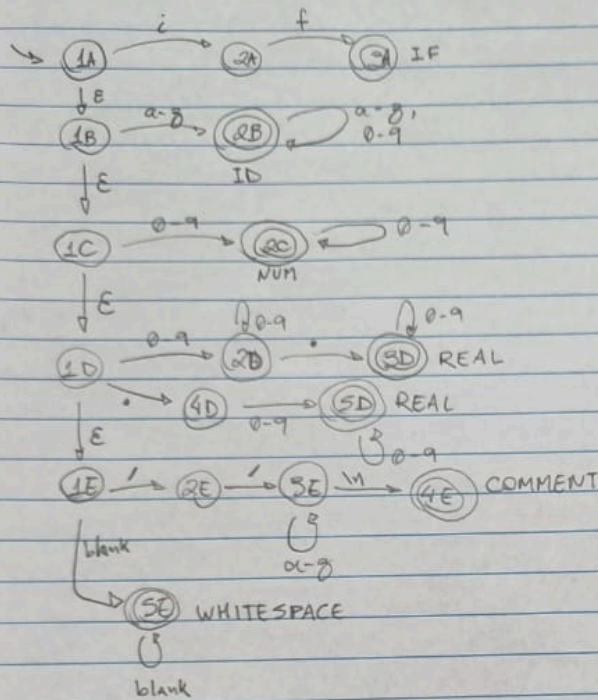
2	x				
* 3	x	x			
* 4	x	x	x		
* 5	x	x	x	x	
* 6	x	x	x	x	x
1	2	3	4	5	

Logo, o autômato já está minimizado

Em conclusão, após análise, vê-se que os autômatos apresentam o mesmo autômato mínimo, diferenciando-se apenas pelos nomes dos estados. Portanto, são equivalentes.

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LISTA 4 - EXERCICIO 4 - LETRA A

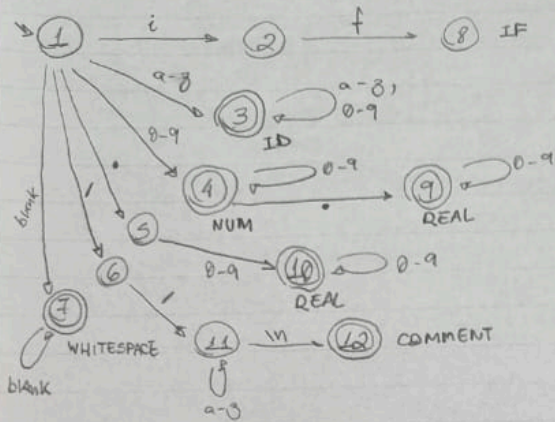


LISTA 4 - EXERCICIO 4 - LETRA B

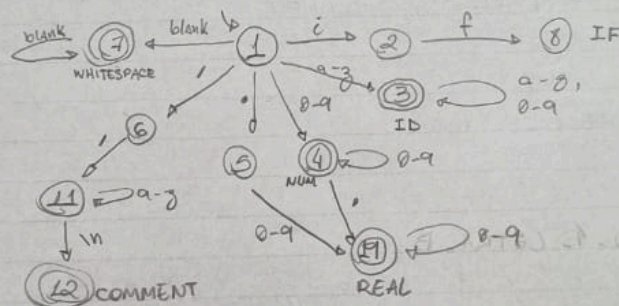
		i	f	a-z	0-9	.	/	\n	blank
1	{1A, 1B, 1C, 1D, 1E}	2A	-	2B	{2C, 2D}	4D	2E	-	5E
2	2A	-	3A	-	-	-	-	-	-
3	*2B	-	-	2B	2B	-	-	-	-
4	{2C, 2D}	-	-	-	{2C, 2D}	3D	-	-	-
5	4D	-	-	-	5D	-	-	-	-
6	2E	-	-	-	-	-	3E	-	-
7	*3E	-	-	-	-	-	-	-	5E
8	*3A	-	-	-	-	-	-	-	-
9	*3D	-	-	-	5D	-	-	-	-
10	*5D	-	-	-	5D	-	-	-	-
11	3E	-	-	3E	-	-	-	-	-
12	*4E	-	-	-	-	-	-	-	-

LISTA 4 - EXERCICIO 4 - LETRA B

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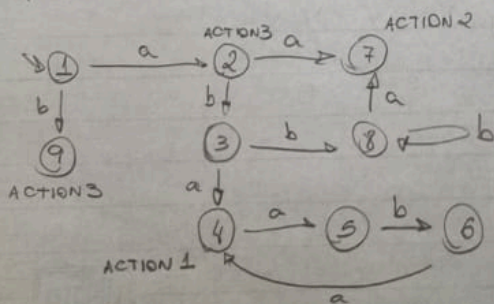


MINIMIZADO:



LISTA 5 - EXERCICIO 3

Autômato:



Análise léxica:

	a	b	
1	2	9	1 $\xrightarrow{a}$ 2 $\xrightarrow{b}$ 3 $\xrightarrow{a}$ 4
2	7	3	"aba" $\Rightarrow$ ACTION 1
3	4	7	
4	5	-	1 $\xrightarrow{a}$ 2 $\xrightarrow{b}$ 3 $\xrightarrow{b}$ 7 $\xrightarrow{a}$ 7
5	-	6	"abba" $\Rightarrow$ ACTION 2
6	4	-	
7	-	-	1 $\xrightarrow{b}$ 9 $\xrightarrow{a}$ 9 $\xrightarrow{b}$ 9 $\Rightarrow$ ACTION 3
8	7	7	1 $\xrightarrow{a}$ 2 $\xrightarrow{a}$ 2 $\xrightarrow{a}$ 2 $\Rightarrow$ ACTION 3
9	-	-	